



BME 4118 Capstone Project Proposal

Topic: A Fluorescence-Based, Proximity-Gated CRISPR-Cas12a Biosensing Platform for Dual-Marker Detection of α -Synuclein on Neuron-Derived Extracellular Vesicles for Early Parkinson's Disease Diagnosis.

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Abstract

Early diagnosis of Parkinson's disease relies on detecting misfolded alpha-synuclein proteins attached to neuronally derived extracellular vesicles. A very tiny fraction of this protein can cross the blood-brain barrier and appear in easily accessible biofluids, such as blood, in high concentrations, since blood is rich in various proteins such as albumins. This project proposes a dual-recognition, enzymatically amplified biosensing platform that couples proximity-based AND-gate DNA logic with CRISPR-Cas12a collateral cleavage to achieve low-background, high-specificity detection of α -synuclein-positive neuronal extracellular vesicles. Two split DNA initiator strands, S1 and S2, each conjugated to a binder aptamer against L1CAM and membrane-associated α -synuclein, respectively, are designed such that simultaneous, spatially proximal binding of both markers on an intact EV surface is required to nucleate a downstream two-hairpin assembly reaction. This hairpin-mediated hybridization reconstitutes a stable, full-length DNA duplex that is recognized as an activator by a Cas12a-crRNA complex, cleaving fluorophore-quencher reporter probes to generate a measurable fluorescence signal. The proposed research will (1) design and construct the dual-binder/hairpin-based AND-gate system, (2) validate its specificity and triggering efficiency in solution and on model membranes, and (3) evaluate the performance of the logic gate on cell-derived extracellular vesicles from Parkinson's disease-relevant models. We expect to receive true positive fluorescence signals only when the L1CAM and the alpha-synuclein protein biomarkers are detected simultaneously through our AND gate DNA logic system.

1. Background

[1]Known as a progressive neurodegenerative disease, Parkinson's disease is caused by loss of dopamine producing neurons in the substantia nigra of the midbrain, due to the misfolding of alpha-synuclein protein at molecular level. Globally, Parkinson's disease is ranked 2nd as the most common form of neurodegenerative disease after Alzheimer's disease. [2]In 2021, the global prevalence of Parkinson's disease was estimated to be greater than 11 million and in 2040, it is estimated to increase by 14 million cases. [3]Movement disorders are symptoms of Parkinson's disease which results in motor dysfunction, since neuron in mid brain containing substantia nigra is affected which controls motor functions. Symptoms such as hand tremor, cognitive disorders such as depression, and cardiovascular diseases are common in parkinson's disease patients. Since it's prevalence is high, early diagnosis is extremely important to aware people to take preventive measures, and such can be done through detecting potential exosome biomarkers of Parkinson's disease, such as clumps of misfolded alpha-synuclein proteins. For diagnosis, screening, prognosis and monitoring of the trend of the disease, exosomal proteins with extracellular nanovesicles have depicted as promising candidate for biomarker detection. Exosomal protein biomarkers of Parkinson's disease are alpha-synuclein, tau, Amyloid β 1-42, and C-X-C motif chemokine ligand 12 (CXCL12) and among them alpha-synuclein is present in highest concentration in Parkinson's disease patients. Recent diagnosis methods are mostly reliable on clinical assessments of motor function, and motor disfunction are visible symptoms after a significant amount of neurodegeration have occurred. Additionally, this disease has a prolonged prodormal phase and depicts non-motor symptoms in patients. So far, no concrete blood test is present to detect this disease and the most challenging part is the specific detection of these high potential protein biomarkers of the pathology in easily accesible biofluid, which is blood. [4]Blood already has a high concentration of different proteins, such as serum albumin, and globulin. The ratio of neuronally derived exosomal vesicles attached the alpha synculein in comparison to these proteins is less, since the extracellular vesicles need to cross the blood brain barrier in central nervous system before appearing in the peripheral blood. Moreover, alpha synuclein protein can also be found in free or soluble form in plasma, which are often red-blood cells derived, and are present in bulk amount, often overshadowing the alpha-synuclein attached to neuronally derived exosomal vesicles from

Parkinson's disease. In serum or plasma, there is heterogeneity in alpha-synuclein quantity, and the outcomes of measurements for diagnostic purposes are not satisfactory. Therefore, our proposed solution, based on fluorescence biosensing mode, is to detect two protein biomarkers, L1CAM and alpha-synuclein, simultaneously on the neuronally derived exosome vesicles using an AND-gate logic approach with activation of the Cas12a system, resulting in fluorescence only if two biomarkers are detected at the same time. By doing so, we are enhancing the specificity of Parkinson's disease biomarker detection and also reducing the effect of free and red blood cell-derived alpha-synuclein in the blood, which would often give false positive results for Parkinson's disease.

2. Literature Review

The review emphasizes how early Parkinson's disease diagnosis from free alpha-synuclein measurements in serum or plasma in bulk shifts to a more specific and sensitive approach, which is detecting neuronally derived extracellular vesicles attached to this biomarker, highlighting some specific limitations of free-protein background, non-neuronal extracellular vesicle contamination, and specific localization of alpha-synuclein on the membrane of extracellular vesicles.

2.1. Alpha-synuclein as a potential biomarker of Parkinson's Disease

[5] Different types of cells, including neurons, can secrete extracellular vesicles, which are membranous nanoparticles and can bind to disease-specific biomarkers. As a neurodegenerative disease biomarker, neuronally derived L1CAM-associated extracellular vesicles are promising and also demonstrate the homeostatic state of nerve cells.[6][7] Misfolded alpha-synuclein aggregates in neurons due to gene mutation are a major biomarker of Parkinson's disease and many other neurodegenerative diseases. Typically, alpha-synuclein is found in the presynaptic terminals of neurons and helps. Structurally, alpha-synuclein is viewed as an intrinsically disordered protein, which means that its 3D structure changes depending on its interaction with other molecules in its biological environment, such as lipids, endogenous small molecules, and proteins, and variation in gene mutations. In Figure 1, the chemical structure of alpha-synuclein is shown, representing 3 functional domains. The N-terminus has 1-60 amino acids, where most of the genetic

mutations happen. The middle region is the non-amyloid- β -component (NAC), which is prone to aggregation by generating β sheets. Lastly, the C-terminus contains anionic residues that allow cationic metal ions, proteins, membrane binding, and small molecules to bind to the protein. [8][9] Parkinson's disease is a result of synucleinopathies, which are the misfolding and aggregation of the proteins into amyloid fibrils, also known as β -sheet fibrils, which makes them insoluble and causes them to aggregate into clumps, forming Lewy bodies inside neurons. As a result, this hampers neurotransmitter (dopamine) release from the presynaptic membrane to the postsynaptic membrane as shown in Figure 2, vesicle recycling and trafficking, mitochondrial function is hampered, and pores can be created in the presynaptic membrane, resulting in excitotoxicity.

2.2. Blood-cell contamination and non-neuronal extracellular vesicles

[10] Although a high concentration of extracellular vesicles can improve the accuracy of α -synuclein measurement, an extracellular vesicle-enriched medium is not necessarily equivalent to a pure neuronal-extracellular vesicle medium. Blood contains abundant non-neuronal extracellular vesicles and various types of proteins, and alpha-synuclein is abundant in red blood cells; hemolysis would cause contamination of the blood with red blood cell-derived alpha-synuclein. In addition, free α -synuclein and α -synuclein aggregates may interact nonspecifically with capture surfaces or detection reagents. These factors create a risk that a positive fluorescence signal is obtained from the presence of α -synuclein somewhere within the assay mixture rather than α -synuclein bound to the membrane of neuronal extracellular vesicles.

2.3. Conventional methods for the detection of alpha-synuclein associated EVs

[11] Conventional measurements of α -synuclein in Parkinson's disease have primarily used ELISA, electrochemiluminescence immunoassays, and digital single-molecule platforms such as Simoa. These methods typically rely on antibody-based capture and detection of total, oligomeric, phosphorylated, or otherwise disease-associated α -synuclein proteins. ELISA is comparatively accessible and easy, but its sensitivity may be insufficient for low-abundance α -synuclein in plasma or extracellular-vesicle containing samples. Electrochemiluminescence platforms

improve analytical sensitivity and multiplexing capacity and have been used to quantify α -synuclein, clusterin, synenin-1, and related markers in L1CAM-immunocaptured extracellular vesicle fractions. Simoa and other digital immunoassays further improve sensitivity by detecting individual immunocomplexes in thousands of femtoliter-scale reaction wells. Measurement of α -synuclein-associated extracellular vesicles in bulk amounts using these methods has provided important evidence that EV-associated α -synuclein may differ between patients with Parkinson's disease and control groups. However, these assays generally measure the total α -synuclein recovered from an extracellular vesicle-rich or immunocaptured medium rather than directly confirming that α -synuclein is associated with an extracellular vesicle, displayed on its membrane, or co-localized with an independent extracellular vesicle biomarker. Consequently, free α -synuclein, protein aggregates, nonspecific adsorption, and non-neuronal particles may contribute to the detection signal, resulting in false positive results.

2.4. Fluorescent and nanomaterial-based biosensors for alpha-synuclein detection

[12]Fluorescent and nanomaterial-based α -synuclein biosensors provide useful supporting technologies for improving recognition sensitivity, conformational selectivity, and optical signal generation, although most have not yet been validated extensively using well-characterized patient-derived extracellular vesicles. Graphene oxide/FRET aptasensors use the fluorescence-quenching ability of graphene oxide and the structural rearrangement of a fluorescently labeled aptamer to detect α -synuclein oligomers.[13] Carbon-dot probes can provide fluorescence enhancement, spectral shifts, or analyte-dependent changes in emission when interacting with biomarker fibrils. Aggregation-induced-emission (AIE) probes have also been investigated because their fluorescence increases when molecular motion is restricted upon binding to α -synuclein aggregation species. These studies have demonstrated highly sensitive readouts from fluorescence biosensing; however, they have not yet solved the problem that the signal comes specifically from the α -synuclein being colocalized with a biomarker of neuronal extracellular vesicles, which is L1CAM.

2.5. Proposed AND-gate strategy for detecting alpha-synuclein.

The central analytical challenge is therefore not simply to increase the fluorescence response to α -synuclein. Rather, the assay must determine whether α -synuclein is associated with the correct particle, in the correct molecular context, and preferably at the exosome membrane surface. L1CAM immunocapture provides a useful enrichment step for neuron-associated extracellular vesicles, but it does not identify whether alpha-synuclein is displayed on the membrane or not. A proximity-dependent DNA logic system offers a complementary solution by requiring simultaneous recognition of α -synuclein and an independent extracellular vesicle membrane marker before Cas12a fluorescence amplification is activated. Such an AND-gated architecture could reduce false signals from free protein, isolated aggregates, and single-marker particles, thereby improving the molecular specificity of α -synuclein detection. Therefore, this study aims to develop a fluorescence biosensing platform that combines L1CAM-based exosome enrichment with dual-marker proximity recognition and Cas12a-mediated signal amplification for selective detection of neuronal extracellular vesicle-associated alpha-synuclein. This project proposes a dual-recognition, enzymatically amplified biosensing platform that couples proximity-based AND-gate DNA logic with CRISPR-Cas12a collateral cleavage to achieve low-background, high-specificity detection of α -synuclein-positive neuronal extracellular vesicles. Two split DNA initiator strands, S1 and S2, each conjugated to a binder aptamer against L1CAM and membrane-associated α -synuclein, respectively, are designed such that simultaneous, spatially proximal binding of both markers on an intact exosome surface is required to nucleate a downstream two-hairpin assembly reaction. This hairpin-mediated hybridization reconstitutes a stable, full-length DNA duplex that is recognized as an activator by a Cas12a-gRNA complex, cleaving fluorophore-quencher reporter probes to generate a measurable fluorescence signal, as shown in Figure 3.

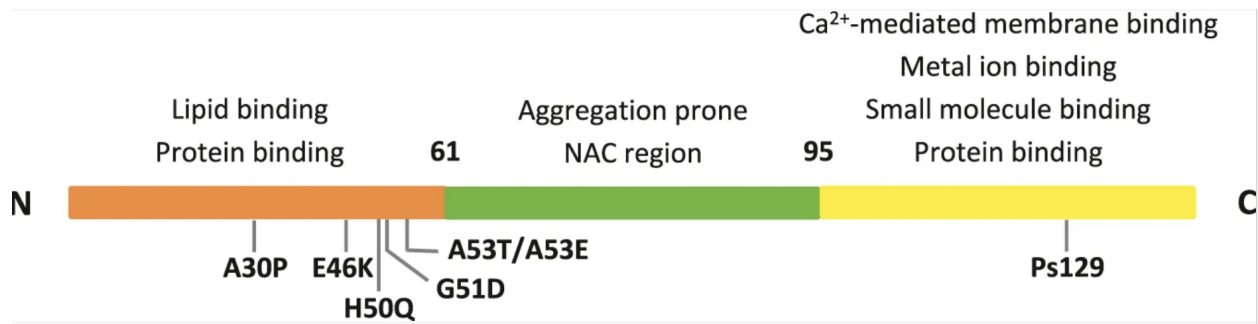


Fig 1: Native chemical structure of alpha-synuclein representing 3 different regions along with aggregation prone region in parkinson's disease.

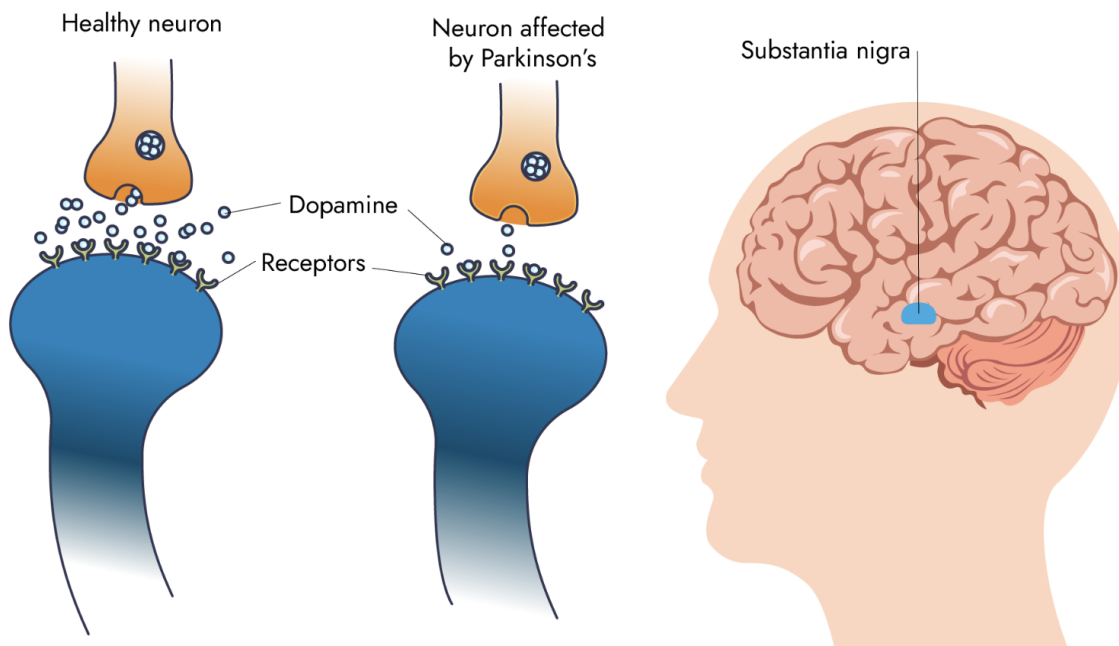


Fig 2: The figure demonstrates the difference in healthy and parkinson's disease synapse.

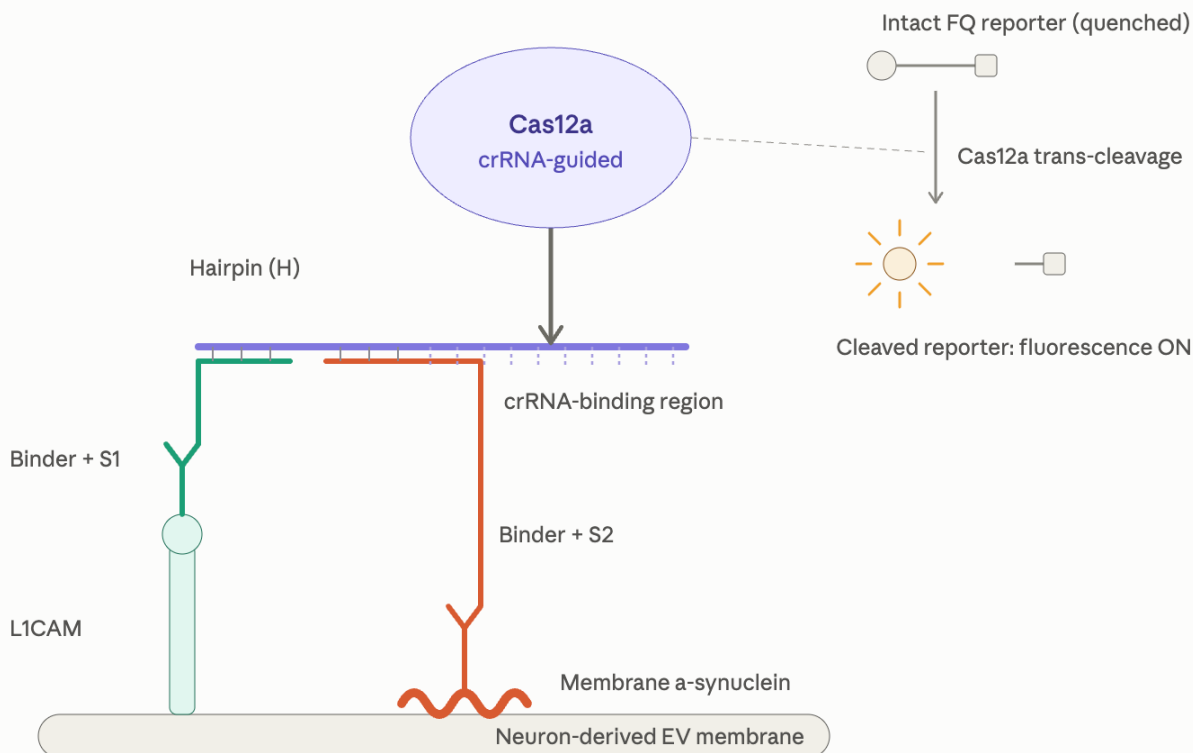


Figure 3: Overview of our proposed dual-marker AND-gate Cas12a fluorescence biosensing platform for detecting neuron-derived, α -synuclein-positive EVs for early Parkinson's Disease Diagnostics.

3.1 Hypothesis

A dual-recognition, proximity-activated AND-gate in which split single-stranded DNA (ssDNA) initiator strands conjugated to binders against L1CAM and membrane-associated α -synuclein reconstitute a complete Cas12a-activating trigger sequence only when both biomarkers are simultaneously present in proximity on the intact surface of a single extracellular vesicle will enable low-background, high-specificity fluorescence detection of neuron-derived, α -synuclein-positive EVs, and will outperform single-marker detection schemes in discriminating disease-relevant EVs from healthy/control EVs.

3.2 Objectives:

1. Design and construct a dual-aptamer proximity logic gate that reconstructs a full DNA initiator strand only when L1CAM and α -synuclein are co-localized on the surface of intact extracellular vesicles (EVs).
2. Validate the specificity and efficiency of the AND-gate in solution and on model membranes, demonstrating that only dual-positive targets trigger initiator formation.
3. Evaluate the performance of the logic gate using cell-derived EVs from Parkinson's disease-relevant models, confirming selective recognition of neuron-derived, α -syn-positive EVs.

4. Design and Methodology

4.1. Cell culture and EV isolation

Wild-type SH-SY5Y cells (L1CAM⁺, low endogenous α -synuclein) and SH-SY5Y cells lentivirally transduced to overexpress A53T α -synuclein (L1CAM⁺/ α -synuclein-high) will be cultured in DMEM with 10% FBS at 37 °C and 5% CO₂. At 70-80% confluency, cells will be switched to exosome-depleted medium for 48 h. The conditioned medium will be cleared by differential centrifugation (300 × g for 10 min, 2,000 × g for 20 min, 10,000 × g for 30 min, 4 °C), filtered through a 0.22 μ m filter, and ultracentrifuged at 100,000 × g for 90 min. The EV pellet will be washed in PBS, re-pelleted, and resuspended in PBS with protease inhibitors. EVs will be characterized by nanoparticle tracking analysis (size and concentration), transmission electron microscopy (morphology), and western blotting (CD9, CD63, CD81, Flotillin-1, L1CAM; calnexin as negative control). A53T overexpression will be confirmed by immunoblotting of cell lysates.

4.2. Design of nanobiosensing probe

An anti-L1CAM antibody or aptamer and an anti- α -synuclein antibody targeting a membrane-accessible epitope (e.g. the LB509 epitope) will be conjugated to split initiator strands S1 and S2 by azide-DBCO click chemistry, using a flexible spacer. Conjugates will be purified by HPLC and verified by PAGE and MALDI-TOF mass spectrometry, and binding affinity will be confirmed by ELISA or surface plasmon

resonance. Strand sequences, junction stem length (4, 6, and 8 nt), and hairpins H1/H2 will be designed and screened in silico with NUPACK. The full-length activator assembled by the hairpins will carry the crRNA-complementary sequence (with a PAM if double-stranded). A FAM/BHQ1-labelled ssDNA reporter will be used for Cas12a readout.

4.3. Feasibility validation of the biosensing mechanism

Activator assembly will be tested in solution with and without a proximity template and analyzed by native gel electrophoresis. Cas12a trans-cleavage will then be monitored by real-time fluorescence in a plate reader, and concentrations of Cas12a, crRNA, hairpins, and reporter will be optimized for signal-to-background ratio. To mimic membrane presentation, recombinant L1CAM and α -synuclein will be immobilized on magnetic beads or liposomes under four conditions: both targets, L1CAM only, α -synuclein only, and neither. Scrambled-DNA and non-binding-antibody controls will be included. A signal only in the dual-target condition will confirm AND-gate behavior, and the limit of detection will be estimated as the blank mean plus 3 SD.

4.4. Validation using extracellular vesicles derived from cell line.

EVs from wild-type and A53T cells will be captured on anti-L1CAM or anti-CD9 magnetic beads, incubated with the binder-DNA conjugates and hairpins, and then with Cas12a/crRNA/reporter. Fluorescence will be measured by plate reader and compared between cell lines after normalizing to NTA particle number, with higher signal expected from A53T EVs. Controls will include EV-depleted medium, beads without capture antibody, single-binder assays, and free recombinant α -synuclein. Proteinase K digestion with and without Triton X-100 will confirm surface-exposed α -synuclein. Results will be cross-validated by Western blot or ELISA, and statistical analysis will use t-tests or one-way ANOVA across at least three independent EV preparations.

5. Budget

Category	Items	Estimated Cost/HKD
Materials and Reagents	Anti-L1CAM and anti-alpha-synuclein antibodies,DNA oligos(S1,S2,hairpins H1/H2),gRNA, Cas 12a enzyme,FQ reporter probe, conjugation/click chemistry kit	2,700
Cell Culture and EV isolation consumables	Culture media, exosome-depleted FBS, culture flasks, filters, ultracentrifuge tubes, protease inhibitors	700
Gel Electrophoresis Consumables	Agarose, DNA ladder, loading dye,running buffer	200
Miscellaneous	Printing, presentation materials	400
Total		4000

Table 1: Estimated budget for reagents and consumables to be used in the research.

Note: This budget assumes that all major high-value items such as fluorescence microscopy and laboratory centrifuges are already available in my supervisor's laboratory. The listed costs therefore cover only consumables and minor accessories. If any of these major devices are not available at the time of purchase, the total project cost will increase accordingly.

6. Schedule

Phases	Key Activities	Timeline (2026/2027)
Phase 1:Design & construct AND-gate	Literature review; design binder-DNA conjugates (S1/S2), hairpins (H1/H2), crRNA; in silico screening (NUPACK); order oligos/antibodies; conjugation chemistry & purification	October-November
Phase 2: Validate specificity & low background	Cell-free activator assembly (gel electrophoresis); Cas12a trans-cleavage optimisation; bead/liposome model-membrane AND-gate testing; single-marker & free-protein background controls	December -January
Phase 3:EV/PD-model evaluation	Culture SH-SY5Y WT & A53T lines; EV isolation & characterisation (NTA, TEM, western blot); apply assay to EVs; compare WT vs A53T signal; specificity controls	February- mid April
Final Report and Presentation	Data analysis; thesis/report writing.	Late April

Word count:2740

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